# Software Architecture

## Scenario Analysis and Classroom Exercise

### Q1. Online Movie Streaming

A movie streaming platform allows users to watch movies using their phones, laptops, and smart TVs. All devices request movie information and streaming services from remote servers.

Which architectural pattern best describes this system? Why?

### Q2. Online Shopping Discount

An online shopping application separates its user interface, discount calculations, and database access. The company frequently changes its discount rules but wants to avoid modifying the user interface or database access code.

Which architectural pattern would you recommend? Why?

### Q3. Student Grade System

A university application displays the same student grades in different ways: students see a grade report, while teachers see charts. When grades change, both displays should reflect the updated information.

Which architectural pattern is most suitable? Identify the responsibilities of its main components.

### Q4. Software Development Tools

A software company uses several tools: a UML editor, a Java code editor, a code generator, and a report generator. All tools need to access and update the same project information without communicating directly with one another.

Which architectural pattern would be most suitable? Explain how these tools share information.

**Q5. Employee Salary Processing**

A company develops a salary processing application. Every month, employee records pass through several steps: checking attendance, calculating salaries, deducting taxes, and generating payslips. Each step processes the output from the previous step.

Which architectural pattern is most appropriate? Why.

**Q6. University Course Registration System**

A university is developing an online course registration system. Students access the system through their browsers to register for courses.

During registration week, thousands of students may access the system simultaneously. The university wants the system to continue working even if one application server fails.

Tasks:

- Draw a simple high-level architecture showing the students' browsers, application servers, and database.
- Suggest one architectural decision to improve the system's availability.
- Explain your decision in one or two sentences.